CLEAR VIEW

Conversation archive & product design brief | November 11, 2026

Archive note: This document consolidates the Clear View design discussion available in this conversation context. Earlier portions of the project were carried forward as a summarized design history, so this is a comprehensive project record, not a guaranteed word-for-word export of every prior message.

1. Product concept

Clear View is a premium bedside glasses dock that also automatically cleans eyewear. The product should first look and feel like an attractive luxury bedside accessory; automatic cleaning is the additional benefit. The design direction is minimalist, futuristic, compact, and visually closer to a premium watch winder than a conventional appliance.

Core user experience: place folded glasses on the top platform; the roof opens; the glasses platform moves into the enclosed chamber; the roof closes; the machine cleans and dries the lenses; a subtle warm glow remains while the glasses are stored; touch the exterior control to retrieve them.

2. Design decisions carried forward

Glasses orientation: glasses are inserted folded, with lenses facing forward. The nose bridge rests in an adjustable cradle, with soft silicone contact points that avoid lens surfaces. Fit must be tested with oversized sunglasses, thick temples, and unusual frame geometries.

Exterior: the current cost-reduced direction is a tinted, hard-coated polycarbonate enclosure rather than multiple smoked tempered-glass panels. Use a structural internal chassis, molded ribs, concealed fasteners, and matte-black details. Earlier premium direction considered smoked glass with black anodized aluminum.

Roof: two opposing panels slide apart into concealed side pockets/tracks. When closed, the panels meet with an overlapping center seam and a suitable gasket to help contain mist. The latest direction uses molded polycarbonate roof panels and integrated guide tracks.

Elevator: a guided platform carries the glasses. The platform needs to stay level, prevent frame movement, and use replaceable silicone contact inserts.

Cleaning: enclosed fine-mist chamber with fixed nozzles aimed at both lens surfaces, a replaceable cleaning-fluid cartridge, and a separate waste collection tray. No microfiber pads should contact lens surfaces.

Drying: centrifugal blower, filtered intake, internal airflow channels, moisture baffle, and concealed rear exhaust. Cleaning and drying airflow modes should be controlled separately.

Controls: weight detection for loading, capacitive touch control for retrieval, position sensing, ambient LED lighting, and safety interlocks. No touchscreen is planned.

3. Updated $50 / 10,000-unit cost target

The user selected one shared motor for the roof and elevator, tinted polycarbonate exterior, and replaceable cleaning-fluid cartridge. The target is $50 in parts per unit at 10,000 units. This is an aggressive engineering target, not a supplier-verified quote.

Part group	Target cost
Molded tinted polycarbonate shell	$8.00
Roof panels, tracks, sliders	$4.00
Elevator platform, guides, inserts	$2.50

Part group	Target cost
Shared geared DC motor	$4.50
Cam drum, gears, belts, linkages	$2.50
Mist pump and nozzles	$4.00
Fluid cartridge and waste tray	$2.00
Centrifugal blower	$3.00
Air filter, ducts, baffle	$1.50
Control board and drivers	$4.00
Sensors and switches	$2.00
LEDs and lighting driver	$0.75
Seals, gaskets, silicone parts	$1.50
Wiring, connectors, hardware	$2.75
Subtotal	$43.00
External power adapter	**$4.50**
Contingency allowance	**$2.50**
TARGET TOTAL	**$50.00**

Cost exclusions: assembly labor, tooling, packaging, freight, duties, development, and testing. The $2.50 contingency is an allowance, not a physical component. All line items require supplier quotes and design validation.

4. Shared-motor mechanical architecture

The proposed mechanism uses one geared DC motor driving a rotating cam drum with coordinated cam tracks. One track controls the roof; another controls the elevator. Mechanical dwell positions allow one movement to stop while the other begins. The electronics use position switches to confirm each indexed state and stop the motor.

Proposed states: A — roof closed/elevator raised; B — roof open/elevator raised; C — roof open/elevator lowered; D — roof closed/elevator lowered for cleaning. Retrieval moves from D to C to B, pauses at B for removal, then returns to A after glasses removal.

A reversible cam alone is not sufficient unless the retrieval-open position can be held while the user removes glasses. The indexed stopping positions and controller are therefore important. The design must prevent roof closure while the elevator is in an unsafe position. Include mechanical stops, torque limiting or current-based obstruction detection, and independent position confirmation.

5. Pneumatic lift idea raised in the discussion

A miniature pneumatic lift was proposed as an alternative to a lead-screw elevator because the moving load is light. Candidate concepts include a silicone bellows, a miniature pneumatic cylinder, or a spring-assisted platform. A bellows could provide quiet movement, but it still needs a pressure source, controlled release, guides, and a way to hold position.

The drying blower should not be assumed capable of powering the lift: centrifugal blowers generally provide airflow at relatively low static pressure. A small diaphragm pump or a mechanically driven air pump may be more suitable, but either adds cost and complexity. A spring-assisted lift may be the lowest-cost option. Pneumatic lift should be compared against the shared-motor cam design before selecting the final architecture.

6. Engineering risks and validation requirements

• Cleaning efficacy: mist and airflow must remove oily fingerprints, not merely wet and dry the lenses.
• Coating compatibility: validate the cleaning solution against anti-reflective, hydrophobic, blue-light, polycarbonate, and plastic lenses.
• Streaking and residue: control droplet size, solution quality, airflow, and drying.
• Folded-frame geometry: verify that folded temples do not shield lens surfaces from nozzles or airflow.
• Moisture management: prevent mist escape, condensation near electronics, and liquid pooling around the elevator.
• Mechanical reliability: test roof alignment, cam wear, lift repeatability, noise, jams, and obstruction protection.
• Maintenance: make the cartridge, waste tray, filter, and cradle accessible for cleaning and replacement.

7. Recommended prototype sequence

1. Prototype roof tracks, elevator guides, cam drum, and shared motor first.
2. Test a separate mist chamber with fixed nozzles and waste collection.
3. Add blower, filter, airflow channels, moisture baffle, and exhaust.
4. Integrate controller, position switches, weight sensing, touch control, and safety interlocks.
5. Finalize the exterior shell after the internal mechanisms and service access are proven.

8. Concept renderings

These are AI-generated visual concept boards, not dimensionally accurate CAD models or validated engineering drawings. The labels, dimensions, and internal arrangements shown in the renderings should be treated as illustrative.

Initial premium concept board



Cost-reduced Rev. 2 concept board

CLEAR VIEW

NEXT GEN – $50 TARGET DESIGN

- AUTO OPEN / CLOSE
- MIST CLEANING
- AIR DRYING
- TOUCH RETRIEVE
- PREMIUM LOOK

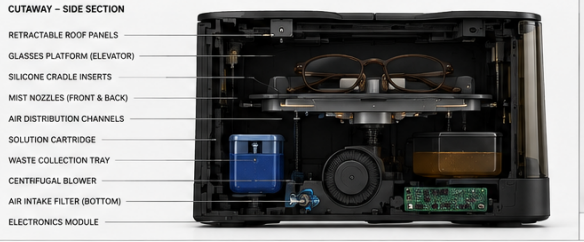


DIMENSIONS (APPROX.)
240 mm (W) x 170 mm (D) x 165 mm (H)

OPERATION SEQUENCE



CUTAWAY – SIDE SECTION



FRONT VIEW – OPEN

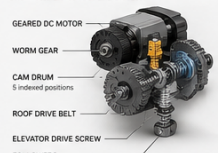


TOP VIEW (OPEN)

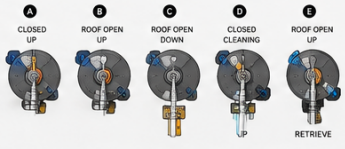


SHARED MOTOR MECHANISM

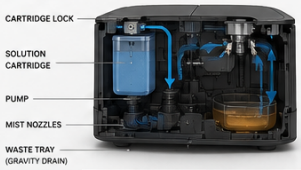
One motor drives roof and elevator



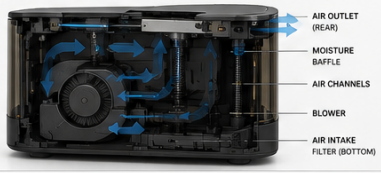
CAM DRUM POSITIONS



FLUID SYSTEM



AIRFLOW SYSTEM



EXTERIOR VIEWS



TOUCH CONTROL



- **TAP ONCE** – Open / Close roof & start cycle
- **TAP AGAIN** – Stop / Pause
- **HOLD 2 SEC** – Reset / Cancel

PARTS LIST & TARGET COST (10,000 UNITS)

PART	TARGET COST
Exterior polycarbonate shell	$8.00
Roof panels, tracks, sliders	$4.00
Elevator platform, guides, inserts	$2.50
Geared DC motor (shared)	$4.50
Cam drum, gears, belts, linkages	$2.50
Mist pump & nozzles	$4.00
Solution cartridge & waste tray	$2.00

PART	TARGET COST
Centrifugal blower	$3.00
Air filter, ducts, baffle	$1.50
Electronics board & drivers	$4.00
Sensors & switches	$2.00
LEDs & lighting	$0.75
Seals, gaskets, silicone parts	$1.50
Wiring, connectors, hardware	$2.75

SUBTOTAL	$43.00
POWER ADAPTER	$4.50
CONTINGENCY	$2.50
TARGET TOTAL	**$50.00**

PARTS ONLY COST TARGET – EXCLUDES ASSEMBLY, TOOLING, PACKAGING, SHIPPING, DUTIES

9. Latest conversation decisions

User-selected Rev. 2 configuration: one shared motor for roof and elevator; polycarbonate exterior; replaceable fluid cartridge; $50 parts target at 10,000 units.

Latest additional idea: investigate air suspension / pneumatic lift for the glasses platform. Compare bellows, pneumatic cylinder, and spring-assisted options against the shared-motor cam lift on cost, quietness, controllability, leakage, and reliability.

Next engineering deliverable: a dimensioned mechanical layout and mechanism study showing the roof tracks, cam drum, elevator guides, drive coupling, chamber, cartridge, waste tray, blower, airflow path, and service access. A real 3D CAD model (STEP/STL) has not yet been created.